

POWIDZ, Poland

WMO: 123360

Lat: 52.379N

Lon: 17.854E

Elev: 371

StdP: 14.50

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.5	11.0	1.0	5.9	7.2	5.9	7.6	12.5	30.6	44.1	26.7	40.6	5.7	60	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	87.6	66.9	84.2	65.6	81.0	64.5	69.5	81.7	68.1	79.8	66.6	76.7	9.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	97.4	72.1	64.4	91.9	71.5	62.9	86.9	70.4	33.8	81.7	32.6	80.3	31.4	76.6	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.9	19.9	17.8	DB	0.0	93.5	7.4	2.7	-5.3	95.5	-9.6	97.0	-13.7	98.5	-19.0	100.5	(4)
			WB	0.4	72.4	7.8	1.8	-5.2	73.6	-9.8	74.7	-14.2	75.7	-19.9	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	28.7	31.7	38.2	48.8	57.1	63.3	67.6	66.6	58.0	48.5	40.1	33.2
	DBStd	15.33	9.86	8.96	8.14	7.62	7.40	6.39	6.03	5.73	5.98	7.05	6.37	8.26
	HDD50	2613	659	512	370	111	19	0	0	0	5	116	300	521
	HDD65	6306	1124	932	831	488	264	106	42	49	224	513	747	986
	CDD50	2092	0	0	5	75	238	399	544	516	245	68	2	0
	CDD65	310	0	0	0	2	19	55	121	100	13	0	0	0
	CDH74	3004	0	0	0	32	209	583	1152	890	137	1	0	0
	CDH80	891	0	0	0	6	40	164	382	264	35	0	0	0
Wind	WSAvg	8.3	9.5	9.0	8.9	8.3	7.9	7.9	7.7	7.3	7.6	8.0	8.7	9.3
Precipitation	PrecAvg	20.8	1.3	1.1	1.4	1.2	2.1	2.4	3.3	2.2	1.7	1.4	1.4	1.4
	PrecMax	28.2	2.7	2.3	2.7	3.1	4.1	4.5	6.0	6.3	3.9	4.6	4.1	3.1
	PrecMin	13.8	0.2	0.2	0.4	0.3	0.7	0.7	0.7	0.6	0.6	0.2	0.0	0.3
	PrecStd	3.3	0.7	0.5	0.6	0.6	0.9	1.1	1.4	1.3	0.9	0.9	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.3	52.8	63.1	78.2	84.5	90.5	92.0	91.6	84.8	71.5	57.5	51.7
		MCWB	47.3	47.3	51.3	58.6	64.4	69.1	68.1	69.6	65.5	60.7	52.4	48.4
	2%	DB	47.0	49.0	58.3	71.6	79.7	84.5	88.0	86.3	77.6	66.4	53.7	48.2
		MCWB	43.9	44.9	48.0	56.3	62.5	65.8	66.5	66.8	62.4	57.7	50.5	46.1
	5%	DB	43.9	45.9	54.0	66.7	75.4	80.7	84.3	82.5	73.4	62.7	51.1	46.1
		MCWB	41.7	42.2	46.7	53.1	60.6	63.9	65.6	65.4	60.4	56.0	48.7	44.1
	10%	DB	41.2	42.9	50.4	62.7	71.2	76.7	80.5	79.0	69.5	58.9	49.0	43.4
		MCWB	39.2	39.8	44.8	51.4	58.4	62.0	64.8	64.0	58.8	54.3	46.9	41.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.6	48.5	53.3	60.7	67.1	71.4	71.2	71.3	66.7	61.7	53.5	48.8
		MCDB	50.4	51.3	60.9	73.3	79.4	85.0	81.9	84.1	81.2	68.7	56.2	50.4
	2%	WB	44.4	45.4	49.9	57.3	64.2	68.2	69.6	69.2	63.9	58.7	51.1	46.4
		MCDB	46.9	48.3	55.4	69.7	76.5	80.2	81.5	81.2	73.7	64.4	53.0	47.7
	5%	WB	41.9	42.5	47.6	54.7	62.0	66.0	68.3	67.7	61.9	56.6	48.9	44.0
		MCDB	43.9	45.7	53.0	64.5	72.3	76.1	79.6	78.8	70.3	61.5	50.8	45.7
	10%	WB	39.3	39.9	45.1	52.3	59.8	63.8	66.8	66.1	60.1	54.7	47.0	41.8
		MCDB	41.2	42.6	50.1	60.6	68.8	72.9	76.8	74.9	67.2	58.7	48.8	43.3
Mean Daily Temperature Range	5% DB	MDBR	8.2	10.4	14.3	18.9	19.2	19.6	19.1	19.2	17.3	13.8	9.0	8.1
		MCDBR	8.9	13.5	21.1	25.1	24.8	25.5	24.9	24.8	23.3	19.1	12.3	9.3
		MCWBR	7.6	10.2	13.3	13.0	10.8	10.7	8.2	9.6	10.6	11.3	9.3	8.0
	5% WB	MCDBR	9.0	13.2	18.9	22.7	21.9	21.9	20.6	20.7	20.2	17.3	11.8	9.2
		MCWBR	7.9	10.5	12.7	12.2	10.6	10.8	8.4	8.9	10.3	10.9	9.6	8.1
Clear-Sky Solar Irradiance	taub	0.314	0.341	0.376	0.421	0.416	0.410	0.439	0.429	0.389	0.367	0.344	0.310	
	taud	2.282	2.282	2.234	2.161	2.228	2.274	2.208	2.241	2.325	2.363	2.336	2.313	
	Ebn at Noon	211	241	255	257	265	268	256	252	249	229	198	192	
	Edh at Noon	20	27	35	43	42	40	42	39	32	25	20	17	
All-Sky Solar Radiation	RadAvg	213	428	805	1318	1630	1706	1587	1402	1008	556	243	154	
	RadStd	34	73	104	171	179	154	191	124	131	79	33	19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	-281	+128	N/A

Nomenclature: See separate page